

Module 6

Sampling

Exercise Sheet

Section A – Basic Problems

Problem 1

Generate a continuous-time sine wave with frequency **5 Hz**.

Sample it at:

- 10 Hz
- 20 Hz
- 50 Hz

Compare the sampled signals.

Problem 2

Illustrate the Sampling Theorem.

Verify that

$$f_s \geq 2f_m \quad \text{or} \quad f_s \geq 2f_m$$

for a given sinusoidal signal.

Problem 3

Demonstrate **under-sampling**, **critical sampling**, and **over-sampling**.

Problem 4

Generate a sampled signal and reconstruct it using interpolation.

Compare the original and reconstructed signals.

Problem 5

Investigate how changing the sampling interval affects the sampled waveform.

Section B – Sampling Concepts

Problem 6

Verify the Nyquist Criterion.

Problem 7

Demonstrate Aliasing.

Problem 8

Compare ideal sampling and practical sampling.

Problem 9

Demonstrate impulse sampling and flat-top sampling.

Problem 10

Observe the frequency spectrum before and after sampling.

Section C – Programming Problems

Problem 11

Implement a simple sampler in Python.

Problem 12

Implement a signal reconstruction algorithm using sinc interpolation.

Problem 13

Add high-frequency noise and recover the signal using a low-pass filter.

Problem 14

Study the effect of different sampling frequencies on FFT.

Problem 15

Estimate the dominant frequency of a sampled signal.

Challenge Problems

Challenge 1

Build an interactive sampling simulator.

Challenge 2

Demonstrate aliasing using different sampling frequencies.

Challenge 3

Compare:

- 8 Hz
- 16 Hz
- 32 Hz
- 64 Hz

sampling of the same signal.

Challenge 4

Reconstruct signals using:

- Linear interpolation
- Cubic interpolation
- Sinc interpolation

Compare the reconstruction quality.

Challenge 5

Study the effect of quantization after sampling.

Challenge 6

Build a mini "Digital Oscilloscope" that displays:

- Original signal
- Sampled signal
- Reconstructed signal
- FFT of the sampled signal